

GENERAL PHARMACOLOGY

MBS230-P&T

UNIT 1.1.1

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Learning Objectives

Students should be able to:

- Define some common terminologies used in Pharmacology and Therapeutics

LS1.1.1- COMMON TERMINOLOGIES USED IN PHARMACOLOGY & THERAPEUTICS

WHAT IS PHARMACOLOGY?

Pharmacology is the science that deals with the study of drugs (Chemicals) or substances and their interactions with living systems:

- Physical and chemical properties of drugs
- Biochemical and physiological effects of drugs
- Knowledge of the history, source, and use of drugs
- Drug absorption, distribution, metabolism and excretion

WHAT IS A DRUG?

Drug: Any chemical substance that affects the physiologic (Biological) processes of a living organism

- A very broad definition of a drug would include “all chemicals other than food that affect living processes”
 - If the effect **helps** the body, the drug is a medicine. However, if a drug causes a **harmful effect** on the body, the drug is a **poison**. The same chemical can be a medicine and a poison depending on conditions of use and the person using it.

COMMON TERMINOLOGIES USED IN PHARMACOLOGY & THERAPEUTICS

Pharmacodynamics: Physiological and biochemical effects of drugs and their mechanism of action at the following:

- Body organ system
- Tissue level
- Cellular level
- subcellular and macromolecular levels

COMMON TERMINOLOGIES USED IN PHARMACOLOGY & THERAPEUTICS

Pharmacokinetics: Refers to movement of the drug in the body and alteration of the drug by the body; includes the processes of:

- absorption
- Distribution
- metabolism
- excretion

Pharmacokinetics also includes terms like; Bioavailability, Bioequivalence, Clearance, Distribution Volume, Half life and Efficacy

COMMON TERMINOLOGIES USED IN PHARMACOLOGY & THERAPEUTICS ... CONT'D

Pharmaco-therapeutics: The application of pharmacological information together with knowledge of the disease for its prevention, mitigation or cure.

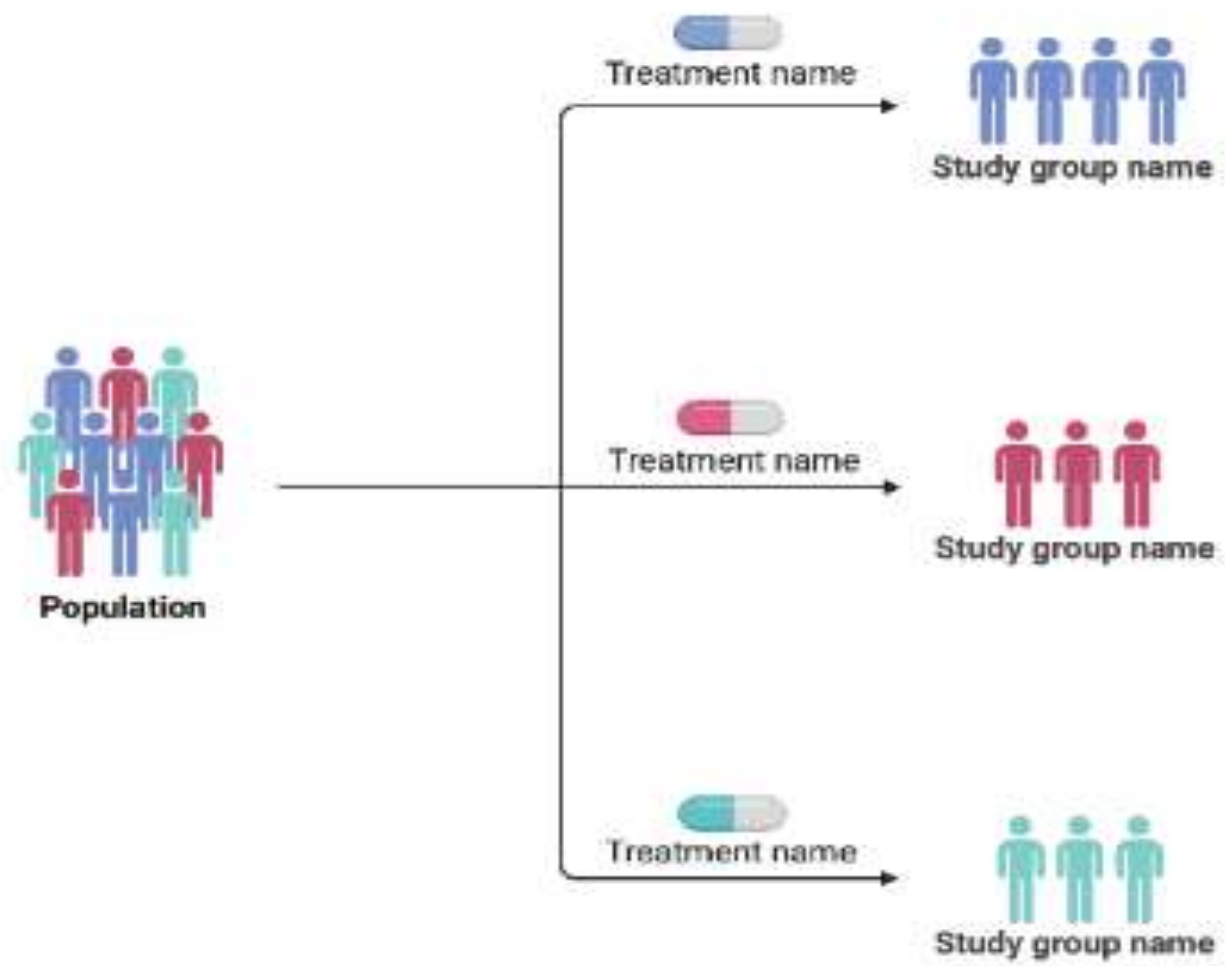
- Selection of the most appropriate drug, dosage and duration of treatment taking into account the specific features of a patient which forms a part of pharmaco-therapeutics.
- Pharmacodynamics also includes; what influences mechanism of drug action, relationship between dose and pharmacological action and dose-effect. (e.g. adverse drug reaction, drug interactions)

COMMON TERMINOLOGIES USED IN PHARMACOLOGY & THERAPEUTICS

Clinical Pharmacology: The scientific study of drugs (both old and new) in man.

It includes **Pharmacodynamics and Pharmacokinetics** investigations in healthy volunteers and in patients; evaluation of efficacy and safety of drugs and comparative trials with other forms of treatment; surveillance of patterns of drug use, adverse effects, etc.

- The aim of clinical pharmacology is to generate data for optimum use of drugs and the practice of 'evidence based medicine'.



COMMON TERMINOLOGIES USED IN PHARMACOLOGY & THERAPEUTICS

Toxicology:

The study of **poisonous effect** of drugs and other chemicals (household, environmental pollutant, industrial, agricultural, homicidal) with emphasis on detection, prevention and treatment of poisonings.

- It also includes the study of adverse effects of drugs, since the same substance can be a drug or a poison, depending on the dose.

COMMON TERMINOLOGIES USED PHARMACOLOGY CONT'D

Chemotherapy:

The treatment of systemic infection/malignancy with specific drugs that have selective toxicity for the infecting organism/malignant cell with no/minimal effects on the host cells.

COMMON TERMINOLOGIES USED PHARMACOLOGY

CONT'D

Minimum Dosage:

The minimum dosage is considered the smallest dosage of the drug that can produce the therapeutic effect.

Maximum Dosage:

The maximum dosage is considered the large dose of the drug that can be safely administered.

Therapeutic effects:

Is a consequence of medical treatment of any kind, results can either be desirable or beneficial

Adverse Effects: Undesirable effects of the drug that may be harmful or serious at doses used for therapeutic effects.

LEARNING OUTCOMES

Students must be able to comprehend:

- Definitions of common terminologies used in Pharmacology & Therapeutics